Series Fault Arc Detection Method for Low-Voltage Electrical Circuits Based on MobileNetV2-WOA

Shuo Xu
Guangzhou Power Supply Bureau
Guangdong Power Grid Co., Ltd
Guangzhou, China
758785014@qq.com

Zhong Xu
Guangzhou Power Supply Bureau
Guangdong Power Grid Co., Ltd
Guangzhou, China
348867958@qq.com

Yu Zhang
Guangzhou Power Supply Bureau
Guangdong Power Grid Co. , Ltd
Guangzhou, China
1139244964@qq.com

Jin Mei Chen
Guangzhou Power Supply Bureau
Guangdong Power Grid Co. , Ltd
Guangzhou, China
1130265190@qq.com

Di Wu*
Guangzhou Power Supply Bureau
Guangdong Power Grid Co. , Ltd
Guangzhou, China
2238169187@qq.com

***Abstract*—To solve the problems of difficult criterion selection and frequent misjudgment in traditional series fault arc detection methods, this study proposes a time-frequency domain feature extraction method and establishes a MobileNetV2-WOA series fault arc detection model. Fault arc current data under different load conditions are collected. Rapid features are extracted using fast Fourier transform and wavelet transform. Detailed features are obtained through a MobileNetV2 model with an attention mechanism. These two types of features are fused to form a comprehensive feature set. A fault detection model is the then constructed based on the fused features. The whale optimization algorithm is introduced. A multi-objective fitness function is adopted to optimize the hyperparameters of the model. Parameter adjustment efficiency and detection accuracy are improved.**

Keywords—series fault arc, time-frequency domain features, feature fusion, whale optimization algorithm, attention mechanism

I. Introduction

With the development of power systems toward intelligence and high efficiency, the safety and reliability of electrical equipment have received increasing attention. Common electrical faults include short circuits, overloads, and poor contacts. Series arcs are generated by poor contact faults, resulting in local overheating and electrical fires [1].

In recent years, detection methods for series fault arcs have been widely studied and developed. In [2], time-frequency domain features of fault current signals were extracted and analyzed to identify fault arcs. However, some extracted features were insufficiently representative, causing misjudgments. In [3-5], voltage drop characteristics caused by fault arcs were analyzed and compared with normal waveforms to identify faults. However, voltage drop features were insufficient to detect accurately fault arcs in complex conditions. Machine learning and deep learning algorithms were studied in [7-9] to improve the accuracy and reliability of fault arc detection. However, computational time was relatively long. In [10-12], multiple feature types were combined to improve fault detection capability. Although accuracy was improved by adding more features, system complexity increased. In [13], The line faults are identified through the algorithm based on convolutional neural networks, which improves the accuracy of identification. In [14-16], random forest is suitable for small samples or nonlinear problems, but it lacks the ability of automatic feature extraction.

Therefore, a low-voltage electrical circuits series fault arc detection method based on MobileNetV2-WOA is proposed in this paper. Fault arc features are extracted by combining rapid and detailed feature extraction methods. The attention mechanism is introduced to fuse features. The whale optimization algorithm (WOA) is used to optimize model hyperparameters. A multi-objective fitness function is adopted to automatically search for optimal parameter settings. A low-voltage circuit series fault arc detection model is finally established.

II. Series fault arc Data Acquisition

This paper builds an experimental platform for low-voltage electrical circuits. The specific structure of the experimental platform is shown in Fig. 1. The experimental platform is mainly composed of a fault arc generator, an air switch, a data acquisition card, a current transformer and various loads. Among them, the fault arc generator is composed of high-temperature resistant alloy electrodes and a precisely controlled arc gap adjustment device. This device simulates series fault arcs of different types of loads by adjusting the spacing between electrodes and the applied voltage.

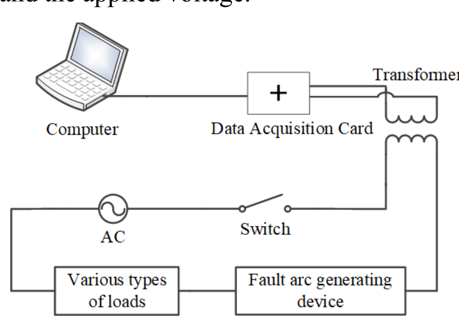

Fig. 1. Experimental platform for low-voltage electrical circuits

This work is support by the Science and Technology Project of China Southern Power Grid Co.Ltd.GDKJXM20240213（030100KC24030013）.

The arc current waveform in the circuit is influenced by the load type. Different load types produce different current waveforms during a Series Fault Arc. Several typical single loads are selected in this study. Resistive, capacitive, and inductive loads are connected to the circuit in sequence. Current waveforms under normal operation and during a Series Fault Arc are collected. This method is used to extract the current features of series fault arcs under different load types.

In this study, the current signals of 10 cycles before and after the occurrence of series fault arc are collected. The sampling duration is 20 ms. The normal and fault current waveforms are shown in Fig. 2.

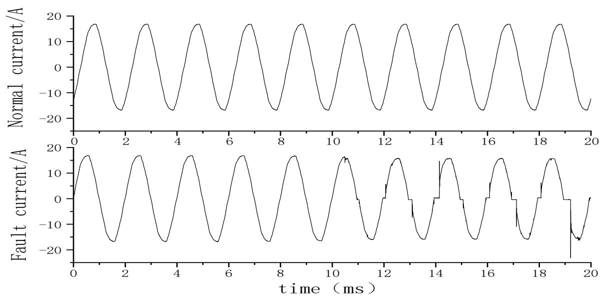

Fig. 2. Normal and fault current waveforms

III. Feature Extraction of Series Fault arcs

The series fault arc current waveform data under different loads are collected. Fast feature extraction is performed on the current data, while detailed featdure extraction is done using the MobileNetV2 model. An attention mechanism is introduced to combine the fast-extracted features with the detailed features, creating a series fault arc feature set. The process is shown in Figure 3.

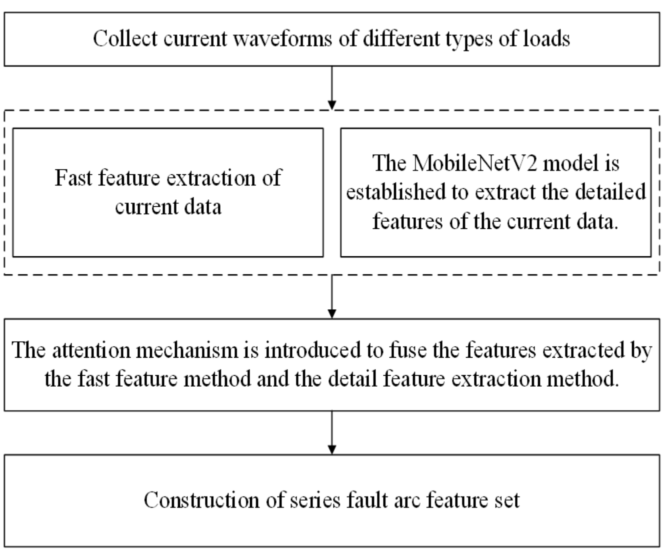

Fig. 3. Process of feature extraction for series fault arcs in low-voltage electrical circuits.

A. *Fast Feature Extraction*

Fast feature extraction is performed using Fast Fourier Transform and Wavelet Transform. Time-domain, frequency-domain, and time-frequency domain features of the fault series arc current under different loads are obtained. In the time domain, five features are selected: peak-to-peak value, average value, kurtosis, root mean square value, and peak factor of the fault series arc current signal.

The peak-to-peak value of the current is the difference between the maximum positive peak and the minimum negative peak of the arc current waveform. It reflects the fluctuation amplitude of the current. The peak-to-peak value is given by

$$I_{P\text{-}P}=I_{max}\text{-}I_{min} \tag{1}$$

Where $I_{max}$is the maximum positive value of the current in the waveform, and I_{min} is the maximum negative value of the current in the waveform.

The average current value represents the mean amplitude of the fault arc current during a specific time period The average current value is given by

$$x_{mean}=\frac{1}{N}\sum_{i=1}^{N} x_i \tag{2}$$

Where N is the number of sampling points within one cycle.

Kurtosis is the ratio of the signal's kurtosis value to the fourth power of the root mean square value. It reflects the smoothness of the current waveform. The kurtosis is given by

$$K_v=\frac{\beta}{x_{rms}^4}=\frac{\frac{1}{N}\sum_{i=1}^{N} x_i^4}{\left(\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}\right)^4} \tag{3}$$

Where K_v is the kurtosis value, x_i is designated as the amplitude of the ith discrete signal, N is specified as the number of signal samples, x_{rms} is established as the root mean square value of the signal, and β is characterized as the normalized fourth-order central moment. The β is given by

$$\beta=\frac{1}{N}\sum_{i=1}^{N} x_i^4 \tag{4}$$

The root mean square value x_{rms} is defined as the square root of the mean square value, which is equivalent to the effective current value. The root mean square value is given by

$$x_{rms}=\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{5}$$

The peak factor measures the distortion of the current signal waveform. It is defined as the ratio of the peak value to the root mean square (RMS) value of the signal. The peak factor is given by

$$C_f=\frac{X_{max}}{X_{rms}}=\frac{max(X_\tau(\tau))}{\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}} \tag{6}$$

The series fault arc frequency-domain features are extracted by selecting nine characteristic values: the amplitude of the fifth harmonic, the amplitude of the seventh harmonic, the amplitude of the ninth harmonic, the harmonic factor of the fifth harmonic, the harmonic factor of the seventh harmonic, the harmonic factor of the ninth harmonic, the total harmonic distortion rate, the mean of the spectrum, and the standard deviation of the spectrum.

The harmonic amplitude refers to the magnitude of the current at different harmonic frequency components.

The harmonic factor is the ratio of the effective value of each harmonic component to the effective value of the fundamental component. The harmonic factor is given by

$$H_k=\frac{D_k}{D_1},k=2,3,\cdots,N/2-1 \tag{7}$$

Where D_k is defined as the amplitude of the kth harmonic component, and D_1 is designated as the fundamental component amplitude.

The Total Harmonic Distortion (THD) measures the total distortion of the current waveform relative to the fundamental co-mponent caused by harmonic components. It reflects the relati-ve strength of non-fundamental components in the signal. TheTotal Harmonic Distortion is given by

$$\mathrm{THD}=\sqrt{\sum_{k=2}^{N/2-1}\left(\frac{D_k}{D_1}\right)^2} \tag{8}$$

Where D_1 is the amplitude of the fundamental wave, and D_k is the amplitude of the k-th harmonic.k represents the harmonic order.

The spectral mean is the average amplitude of the current signal over ten cycles. The spectral mean is given by

$$X_{mean}=\frac{1}{N}\sum_{i=1}^{N} X_i \tag{9}$$

The spectral standard deviation is the square root of the sq-uared difference between the amplitude of the discrete signal o-ver one cycle and its spectral mean. The spectral standard devi-ation is given by

$$X_{std}=\sqrt{\frac{1}{N}\sum_{i=1}^{N} (X_i-X_{mean})^2} \tag{10}$$

The time-frequency domain features of the fault series arc c-urrent signal are extracted. Wavelet transform is applied to the collected fault series arc signal for wavelet decomposition, ob-taining detail coefficients in different frequency bands for time-frequency domain feature extraction. The discrete wavelet tra-nsform is given by

$$\mathrm{DWT}(m,n)=\frac{1}{\sqrt{2m}}\int_{-\infty}^{+\infty} f(t)\psi^*\left(\frac{t-2^{mn}}{2^m}\right)dt,\ \ m,n\in Z \tag{11}$$

The db4 wavelet is used to perform a four-level wavelet tr-ansform on the fault series arc signal. The waveform of the det-ail coefficients after transformation is shown in Fig 4. Through analysis, the detail coefficients at the D4 level effectively refle-ct the variation characteristics of the arc current. Therefore, in this study, the mean value and variance of the D4 detail coeffi-cients are selected as time-frequency domain features.

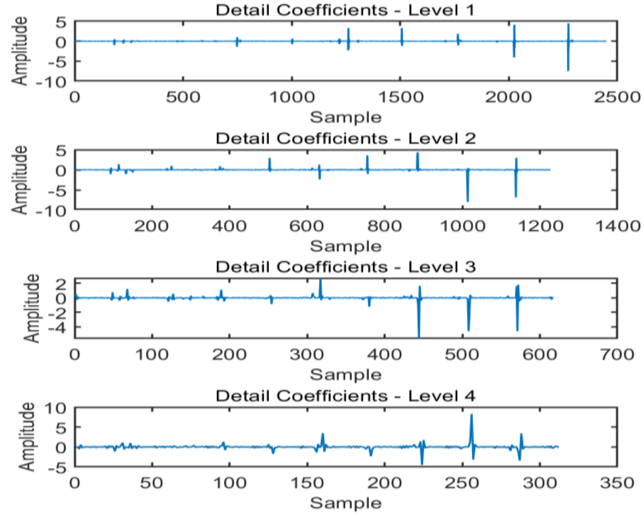

Fig 4 Waveforms of Detail Coefficients After Wavelet Transform

B. *Detail Feature Extraction*

This study proposes a MobileNetV2-optimized feature extr-action framework for series fault arcs, integrating inverted resi-dual structures, linear bottlenecks, and depthwise separable con-volutions to enhance detailed characteristic capture.

The inverted residual structure includes three steps: expans-ion, extraction, and compression. It adjusts the number of cha-nnels to keep useful features while reducing computation. Lin-ear bottleneck modules retain essenti-al information by limiting the final nonlinear activations. Depthwise separable convoluti-on, used as the core operator, splits st-andard convolution into depthwise and pointwise stages. This de-sign models channel correlations through adaptive kernels, balancing representation power and efficiency.

When extracting the detailed features of the series fault arc, the wavelet transform is first used to convert the series fault arc data into a time-frequency image. These images are then resize-d to meet the fixed input size required by MobileNetV2. Thro-ugh (12), the pixel value is mapped from [0,255] to t-he range of [0,1], and the pixel value is normalized. In order to enhan-ce data diversity and prevent overfitting, data enhancement tec-hniques such as random rotation and scaling are used.

$$X_{normalized}=\frac{X_{input}}{255} \tag{12}$$

Based on (13) , Multiple convolutional layers are used to ex-tract both low-level and high-level features from the image. T-he initial convol-utional layer applies a 3x3 kernel and reduces the image size th-rough a stride-2 convolution while extracting preliminary features.

$$X_1=Conv3x3\left(X'_{input},W_1,b_1\right) \tag{13}$$

Where X'_{input} is the image obtained after the normalized im-age is augmented using techniques such as random rotation and scaling, W_1 and , b_1 are designated as the convolution kernel a-nd bias respectively, with X_1 being characterized as the post c-onvolution feature map.

Based on (14), the core operation of MobileNetV2 is the inverted residual module.In each inverted residual module, depthwise convoluti-on is first performed.

$$X_{bottleneck}=DepthwiseConv\left(X_1,W_{depthwise},b_{depthwise}\right) \quad (14)$$

Then, based on (15) , feature fusion between channels is p-erformed using expanded pointwise convolution.

$$X_{expanded}=Conv1x1\left(X_{bottleneck},W_{expand},b_{expand}\right) \quad (15)$$

Finally, based on (16), a linear bottleneck is applied to further optimize feature representation and avoid nonlinear loss.

$$X_{final}=LinearBottleneck\left(X_{expanded},W_{bottleneck},b_{bottleneck}\right) \quad (16)$$

The feature map $X_{expanded}$ is processed through extended c-onvolution; the weight $W_{bottleneck}$ and bias $b_{bottleneck}$ are assigne-d to the linear bottleneck layer respectively; the output feature map X_{final} from this layer contains arc distribution patterns, m-orphological changes, and abnormal current fluctuations as detai-led features.

C. Construct Fusion Feature Set ba sed on Attention Mechanism

The low-level features captured by MobileNetV2 primarily represent local spatial structures, allowing the recognition of edges, amplitude changes, and local pattern variations in the signal. High-level features, after deep nonlinear transformations, represent abstract category information, demonstrating stronger discriminative power and generalization ability.

Based on (17) , in the fusion stage, the fast-extracted physical features are weighted and fused with the deep features extracted by MobileNetV2.

$$X_{fused}=\alpha \cdot X_{manual}+(1-\alpha) \cdot X_{mobile} \quad (17)$$

Where α is defined as the weighting parameter, X_{manual} is d-esignated as the rapidly extracted features, X_{mobile} is characteri-zed as the deep features obtained through MobileNetV2, and X_{fused} is established as the fused features.

On this basis, based on (18) , an attention mechanism is introduced using the Squeeze-and-Excitation (SE) module to weight the features of each channel. The SE module first extracts global features of each channel through global average pooling.

$$z_c=GlobalAveragePooling\left(X_{fused},c\right) \quad (18)$$

Based on (19) , the importance of each channel is calculated through a two-layer fully connected network.

$$s_c=\sigma(W_2 \cdot \delta(W_1 \cdot z_c+b_1)+b_2) \quad (19)$$

Where σ is defined as the Sigmoid activation function, δ is designated as the ReLU activation function, and W_1,W_2,b_1, and b_2 are characterized as trainable parameters. Finally, based on 20 , the features after attention weighting are recalibrated.

$$X_{fused}^{'}=s_c \cdot X_{fused,c} \quad (20)$$

In (20), $X_{fused}^{'}$ represents the weighted fusion feature.

IV. Low-Voltage Circuits Series fault arc Detection Model Based on MobileNetV2-WOA

Real-time constraints in series fault arc detection limit the applicability of conventional CNNs. This prompts the development of a MobileNetV2-WOA enhanced framework, which uses lightweight depthwise convolutions for low-voltage electrical circuits series fault arc detection.

During training, the Whale Optimization Algorithm (WOA) optimizes the hyperparameters of MobileNetV2 through global search mechanisms inspired by cetacean foraging behavior. This enhances arc fault detection accuracy and operational efficiency. Implementation steps include:

(1) Population and Parameter Initialization: An initial population of N individuals is randomly generated. Each individual represents a set of hyperparameters, including width multiplier, resolution multiplier, and learning rate.The maximum number of iterations T_{max} and the convergence threshold are set.

(2) Fitness Evaluation: Based on (21) , the fitness function is defined as a weighted summation of the false negative rate (FNR) and false alarm rate (FAR).

$$Fitness=\alpha \bullet FNR+\beta \bullet FAR \text{（}\alpha+\beta=1\text{）} \quad (21)$$

Where the search range is controlled by parameter α, and the shape of the spiral update path is regulated by parameter β.

(3) Position Update: Global search and local exploitation phases are dynamically balanced through mathematical models. The update process is divided into three methods.

1) Based on (22) and (23), the prey is surrounded. The condition is that the probability p < 0.5 and the coefficient | A | < 1.

$$D=\left|C \bullet X^*\text{（t）}-X\text{（t）}\right| \quad (22)$$

$$X\text{（t+1）}=X^*\text{（t）}-A \bullet D \quad (23)$$

The algorithm defines D as the distance vector between the current optimal solution and target prey, t as the iteration count,X^*（t）and X（t）as the position vectors of the optimal solut-ion and current whale location, respectively, and X（t+1） as t-he updated position. Coefficient vectorsAandCgovern search dynamics, A=2a*r-a ,C=2*r,a=2-$\frac{2t}{t_{max}}$,while the convergence fact-or a linearly decays from 2 to 0 with increasing iterations. A uniformly distributed random vector $r \in [0,1]$ introduces stochast-ic exploration.

2) The bubble-net predation mechanism employs two principal strategies: contraction encircling and spiral position updating. As defined in (24), the coefficient vector derives from convergence factor a, which randomly fluctuates within [-a, a]. When condition $|A| \leq 1$ is satisfied, individual whales execute position updates through contraction encircling.

$$X(t+1)=D'*e^{bl}*\cos(2\pi l)+X*(t) \quad (24)$$

The vector D'=|X*(t)-X(t)| represents the distance between the individual whale agent and the target prey. The parameter "b" denotes the logarithmic spiral constant, a predefined constant with random values uniformly distributed in the interval [-1,1]. To simulate both the spiral upward movement and the progressive encirclement contraction, based on (25), the Whale Optimization Algorithm (WOA) uses a probabilistic mechanism that alternates between the shrinking encircling strategy and spiral position updating, each with a 50% probability.

$$X(t+1)=\begin{cases} X^{*}(t)-A*D, p<0.5 \\ X^{*}(t)+D'*e^{bl}*\cos(2\pi l), p\geq 0.5 \end{cases} \tag{25}$$

Where p is a random number within the range [0,1].

3) Based on (26) and (27), whales locate prey based on a variable vector A.When A>1,whales adopt a random strategy to select updated individuals rather than using optimal solutions to update their positions. This random search strategy enhances the global exploration capability of the algorithm.

$$X(t+1)=X_{road}(t)-A*D \tag{26}$$

$$D=|C*X_{road}(t)-X(t)| \tag{27}$$

Where $X_{road}(t)$ is defined as the position vector of randomly selected individuals within the population. In summary, the basic whale optimization algorithm is constituted through the integration of three search mechanisms.

(4) The optimal hyperparameter combination $X^{*}(t)$ is output when the maximum iteration count T_{max} is reached or when fitness variation falls below a threshold. Under this combination, MobileNetV2 achieves good convergence on both the training and validation sets. It balances accuracy and generalization, mitigating overfitting and underfitting, and constructs the final model. The model is then tested using the arc feature set.

V. Model Performance Verification

This study uses a confusion matrix to validate model performance based on four core metrics: accuracy, recall, precision, and specificity. The definitions of these metrics are provided in Table I.

TABLE I. Confusion Matrix Indicator Definitions

confusion matrix	*PredictedFault (Positive)*	*Predicted Normal (Negative)*	*Total*
True Fault	TP(Normal)	FN(Missed Detection)	TP+FN
True Normal	FP(False Alarm)	TN(Normal)	FP+TN
Total	TP+TN+FP+FN		

The accuracy, recall, precision, and specificity are given by

$$\text{Accuracy}=\frac{TP+TN}{TN+FN+FP+TP} \tag{28}$$

$$\text{Recall}=\frac{TP}{TP+FP} \tag{29}$$

$$\text{Precision}=\frac{TP}{TP+FN} \tag{30}$$

$$\text{Specificity}=\frac{TN}{TN+FP} \tag{31}$$

In order to illustrate the detection ability of the proposed series arc fault detection method for low-voltage electrical circuits based on MobileNetV2-WOA, comparative experiments were carried out with convolutional neural networks (CNN) and BP neural networks. The same feature values defined in this study were used by all three methods for performance evaluation. The training set was composed of 100 samples each from resistive, capacitive, and inductive fault arc data. These samples were used for feature extraction. The test set included 3,000 samples, with 80% fault arc data and 20% normal current data. The final test results are shown in Table II.

TABLE II. Comparison of Various Model Test Indicators

Algorithm	*Accuracy rate/%*	*Recall rate/%*	*Precision rate/%*	*Specificity/%*
Convolutional neural network model	94.67	97.67	96.13	97.00
BP neural network model	97.33	97.83	95.93	98.21
MOBILENETV2-WOA model	99.33	98.83	98.74	99.60

In order to better demonstrate the detection ability of the method type in this paper for mixed loads, it is compared with the convolutional neural Network (CNN) and the BP neural network. The training data of the models constructed based on these three algorithms are all tested for performance using the feature set extracted in this paper. Table II shows that the proposed method outperforms convolutional neural networks and BP neural networks in accuracy, recall, and precision. This validates the superior performance of the model in series fault arc detection.

In order to demonstrate the detection ability of the method proposed in this paper for different loads, the accuracy rate, recall rate and precision rate of different methods for different loads were calculated. The method used in this paper has better accuracy rate, recall rate and precision rate compared with convolutional neural network and BP neural network, and can better conduct series fault arc detection. The results are shown in Table III、IVand V.

TABLE III. The Accuracy Rates of the Three Methods for Different Loads

Algorithm	*Convolutional neural network model*	*BP neural network model*	*MOBILENETV2-WOA model*
Resistive load	84	91	96
Capacitive load	96	98	99
Inductive load	95	100	100

TABLE IV. THE RECALL RATES OF THREE METHODS FOR DIFFERENT LOADS

Algorithm	*Convolutional neural network model*	*BP neural network model*	*MOBILENETV2-WOA model*
Resistive load	80	85	95
Capacitive load	96	98	100
Inductive load	98	100	100

TABLE V. THE PRECISION RATES OF THREE METHODS FOR DIFFERENT LOADS

Algorithm	*Convolutional neural network model*	*BP neural network model*	*MOBILENETV2-WOA model*
Resistive load	93	99	99
Capacitive load	96	96	98
Inductive load	93	100	100

VI. CONCLUSION

1) This study uses a combined method for rapid and detailed feature extraction. It reduces computational complexity and improves extraction efficiency compared to time-domain, frequency-domain, time-frequency domain feature extraction methods, and traditional CNN-based adaptive feature extraction approaches.

2) In this study, the MobileNetV2 neural network model is used to automatically extract detailed features from the current signal. Then, the SE Attention mechanism fuses these features to remove redundant information and retain the most discriminative ones.

3) The proposed MobileNetV2-WOA-based low-voltage elect-rical circuits series fault arc detection model outperforms conv-olutional neural networks and BP neural networks in se-ries fa-ult detection. It demonstrates practical utility in the field of ser-ies fault arc detection.

REFERENCES

[1] J. He, "Research on Low-Voltage AC Series Arc Fault Detection Method Based on Time-Frequency Domain Characteristic Analysis," Harbin Institute of Technology, 2023..

[2] Liu, Y.-J. Chen, C.-I.; Fu, W.-C.; Lee, Y.-D.; Cheng, C.-C.; Chen, Y.-F. A Hybrid Approach for Low-Voltage AC Series Arc Fault Detection. Energies 2023, 16, 1256.

[3] W. Wang, B. Xu, G. Zou et al., "A Series Arc Fault Detection Method for Low-Voltage AC Based on Voltage Characteristic Energy," Power System Protection and Control, vol. 51, 19, pp. 81-93, 2023.

[4] Q. Lu, Z. Ye, M. Su, Y. Li, Y. Sun and H. Huang, "A DC Series Arc Fault Detection Method Using Line Current and Supply Voltage," in IEEE Access, vol. 8, pp. 10134-10146, 2020,

[5] W. Hu, F. Yang, Y. Lei et al., "Fault Series Arc Detection Method Based on Voltage Sag Characteristics and Hausdorff Distance Algorithm," Power Supply and Utilization, vol. 41, pp. 64-74, 2024.

[6] Ke, Y.; Zhang, W.; Suo, C.; Wang, Y.; Ren, Y. Research on Low-Voltage AC Series Arc-Fault Detection Method Based on Electromagnetic Radiation Characteristics. Energies 2022, 15, 1829.

[7] K. Wang, X. Lin, F. Wei and H. Wei, "Detection of Series Arcs in Low-Voltage AC Power Lines Using Load Side Voltage Sag and Waveform Similarity Comparison," in IEEE Access, vol. 12, pp. 1483-1496, 2024,

[8] Dang, H.-L.; Kwak, S.; Choi, S. Various Feature-Based Series Direct Current Arc Fault DetectionMethods Using Intelligence Learning Models and Diverse Domain Exclusion Techniques. Machin 2024.

[9] Q. Yu, J. Xu and Y. Yang, "Complex Branch Fault Arc Detection Based on CNN-LSTM Model," Journal of Safety Science and Technology, vol. 18, pp. 204-210, 2022.

[10] B. Li and J. Wu, "Low-Voltage Arc Fault Identification Using a Hybrid Method Based on Improved Salp Swarm Algorithm‑Variational Mode Decomposition‑ Random Forest," in IEEE Access, vol. 12, pp. 15410-15418, 2024.

[11] L. Cai, Z. Wang, J. Li et al., "Intelligent Detection Technology for Transformer Faults Based on Improved Random Forest Algorithm," Adhesion, vol. 51, no. 10, pp. 153-156, 2024.

[12] X. Liu, D. Huang, T. Jing and Y. Zhang, "Detection of AC Arc Faults of Aviation Cables Based on H-I-W Three-Dimensional Features and CNN-LSTM Neural Network," in IEEE Access, vol. 10, pp. 106958-106971, 2022.

[13] Syifaul Fuada, Hasbi Ash Shiddieqy, Trio Adiono, "A High-Accuracy of Transmission Line Faults (TLFs) Classification based on Convolutional Neural Network," in INTL JOURNAL OF ELECTRONICS AND TELECOMMUNICATIONS, 2020, VOL. 66, NO. 4, PP. 655-664.

[14] B. Li and J. Wu, "Low-Voltage Arc Fault Identification Using a Hybrid Method Based on Improved Salp Swarm Algorithm–Variational Mode Decomposition– Random Forest," in IEEE Access, vol. 12, pp. 15410-15418, 2024.

[15] Dowalla, K.; Bilski, P.; Łukaszewski, R.; Wójcik, A.; Kowalik, R. A Novel Method for Detection and Location of Series Arc Fault for Non-Intrusive Load Monitoring. Energies 2023, 16,171.

[16] Cai Li, Wang Zheng, Li Jingya, et al. Research on transformer fault intelligent detection technology based on improved random forest algorithm [J].bonding, 2024,51 (10) : 153-156.